Michael Amir

ma2151@cst.cam.ac.uk • michaelamir.com

Research Profile

As AI models are increasingly deployed on physical devices, from service robots to autonomous vehicles, humans need ways to verify which models are active, what they are trying to do, and whether their behavior is safe and robust. This is critical for trust, adoption, liability, and preventing misuse. My research addresses the fundamental question: **How can we remotely verify the provenance, intent, and safety of embodied agents?**

- **Policy Provenance and Watermarking:** How can we verify ownership and detect misuse of neural policies deployed on physical robots, using only remote observation data such as video footage?
- **Auditing Intent in Physical AI:** How can autonomous systems signal what they're trying to do in a way that's remotely verifiable, for accountability, trust, and accident investigation?
- **Guarantees for Neural Policies:** Neural policies are opaque and come with no behavioral guarantees, making them difficult to trust. How can we deploy these networks while preserving formal guarantees about their safety and other behavioral desiderata?

Professional Experience

Prorok Lab at University of Cambridge — **Cambridge, UK**
Research Associate — *2024–*

- **Remotely Detectable Robot Policy Watermarking:** The growing use of powerful, IP-sensitive robotics models (incl. foundation models and driving policies) creates a critical need to verify ownership and detect unsafe misuse. I develop spectral coherency-based methods for remotely identifying which policy is deployed on a physical robot at any given time using only seconds of video footage, creating an audit trail for neural policies without affecting performance (ICLR 2026).
- **Motion-based Steganography for Physical AI:** Beyond provenance, it is important to verify the intent behind AI output. Building on my work on watermarking, I develop steganographic methods to embed verifiable "intent metadata" into robot motion (e.g., what mission a service robot is performing and who requested it), which enables remote auditing of what autonomous systems are trying to do. This is important for accident investigation, accountability, and user trust in Physical AI.
- **Guarantees for Neural Policies via Hybrid Methods:** Pure neural policies are effective but error-prone and have no safety guarantees. I work on hybrid architectures where neural networks help improve, rather than replace, systems with formal guarantees. I show that policies are safer, and more robust to perturbations or challenging OOD scenarios, when used together with classical optimization methods (via co-design). In multi-agent settings, I also show that including graph neural network heuristics within search improves performance while guaranteeing task completion (AAAI 2026 Oral, CoRL 2025).
- **Mentorship:** Mentored 7 graduate students on topics in AI provenance, robustness, safety, and multi-agent RL.

Technion — **Haifa, Israel**
Postdoctoral Fellow — *2023*

- **Robustness in Agent Swarms:** I studied how agent failures and adversarial conditions impact the robustness and resilience of distributed robotic systems. This established a class of algorithms that enable agents to complete their mission while remaining highly adaptive to crashes and other unexpected, dynamic conditions (IEEE T-RO, AAMAS, RSS).

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Research Collaboration — *2022–2023*

- Developed graph-based statistical learning methods for high-dimensional metagenomic data, which achieved SOTA accuracy in predicting inflammation markers.

Education

Technion **Haifa, Israel**
PhD in Computer Science, GPA: 98.7 *2018–2023*
Thesis: "Multi-A(ge)nt Systems on Graphs" · Advised by Prof. Alfred M. Bruckstein
Focus: Resilient Multi-Agent Algorithms, Graph Theory, and Stochastic Control.

Technion **Haifa, Israel**
MSc. in Computer Science, Thesis grade: 96 *2015–2017*
Thesis: "Probabilistic Pursuits on Graphs" · Advised by Prof. Alfred M. Bruckstein

The Open University **Israel**
B.A. in Mathematics and Philosophy, Magna cum Laude *2009–2015*
Young Students Excellence Program (Completed concurrent with secondary education)

Awards

2024-2026: Postdoctoral Affiliate of Trinity College, University of Cambridge

2022: Runner-up for Rothschild Fellowship

2021: National Excellence Award in Smart Transportation Research

2020: Research Excellence Award (Technion)

2020: Best Research Poster Award (Technion)

2018–2023: Technion Excellence Scholarship

Mentorship

Past **Samuel Ratnam**: *Inverse Constitutional AI for Identifying Divergent Behavior in LLMs.* (ERA Fellowship)

Hao Xiang Li: *Scaling Co-design with Diffusion Models.* (Part III Thesis, Cambridge)

Tali Motzkin: *Covert Steering of Multi-Agent Systems via Hypergraph Neural Networks.* (MPhil Thesis)

Alon Shats: *Ant-like Competitive Coverage.* (MSc., Bar-Ilan University)

Current Mario Gomez Andreu · Lorenzo Magnino · Benjamin Chang (Topics: Watermarking, Robustness, Co-design)

Publications

Journal Articles...

Michael Amir and Alfred M. Bruckstein. "Time, Travel, and Energy in the Uniform Dispersion Problem". *IEEE Transactions on Robotics* (2025).

Dmitry Rabinovich*, **Michael Amir***, and Alfred M. Bruckstein. "Optimally Reordering Mobile Agents on Parallel Rows". *Theoretical Computer Science* 985 (2024).

Michael Amir, Noa Agmon, and Alfred M. Bruckstein. "A Locust-Inspired Model of Collective Marching on Rings". *Entropy* 24.7 (2022), 918.

Michael Amir and Alfred M. Bruckstein. "Probabilistic pursuits on graphs". *Theoretical Computer Science* 795 (2020), 459–477.

Conference Proceedings...

Rishabh Jain, Keisuke Okumura, **Michael Amir**, and Amanda Prorok. "Graph Attention-Guided Search for Dense Multi-Agent Pathfinding". **Oral Paper** at AAAI 2026. arXiv: `2510.17382 [cs.AI]`.

Michael Amir*, Matteo Bettini*, and Amanda Prorok. "When Is Diversity Rewarded in Cooperative Multi-Agent Learning?". *Accepted to ICLR 2026*. arXiv: `2506.09434 [cs.MA]`.

Michael Amir*, Manon Flageat*, and Amanda Prorok. "Remotely Detectable Robot Policy Watermarking". *Accepted to ICLR 2026*. arXiv: `2512.15379 [cs.RO]`.

Rishabh Jain, Keisuke Okumura, **Michael Amir**, Pietro Lio, and Amanda Prorok. "Pairwise is Not Enough: Hypergraph Neural Networks for Multi-Agent Pathfinding". *Accepted to ICLR 2026.*

Hao Xiang Li, **Michael Amir**, and Amanda Prorok. "Scaling Multi-Agent Environment Co-Design with Diffusion Models". *Accepted to ICML 2026*. arXiv: 2511.03100 [cs.LG].

Michael Amir, Guang Yang, Zhan Gao, Keisuke Okumura, Heedo Woo, and Amanda Prorok. "ReCoDe: Reinforcement Learning-based Dynamic Constraint Design for Multi-Agent Coordination". Proceedings of 2025 Conference on Robot Learning (CoRL2025). 2025. arXiv: 2507.19151 [cs.RO].

Michael Amir*, Dmitry Rabinovich*, and Alfred M. Bruckstein. "Patrolling Grids with a Bit of Memory". *Proc. of the Intl. Workshop on the Algorithmic Foundations of Robotics (WAFR)*. Chicago, USA, 2024.

Michael Amir, Yigal Koifman, Yakov Bloch, Ariel Barel, and Alfred M. Bruckstein. "Multi-Agent Distributed and Decentralized Geometric Task Allocation". *Proc. of the IEEE Conference on Decision and Control (CDC)*. 2023.

Ori Rappel*, **Michael Amir***, and Alfred M. Bruckstein. "Stigmergy-based, Dual-Layer Coverage of Unknown Indoor Regions". *Proc. of the Intl. Conf. on Autonomous Agents and MultiAgent Systems (AAMAS)*. London, United Kingdom, 2023.

Alon Shats, **Michael Amir**, and Noa Agmon. "Competitive Ant Coverage: The Value of Pursuit". *Proc. of the IEEE/RSJ Intl. Conf. on Intelligent Robots and Systems (IROS)*. Detroit, United States, 2023.

Michael Amir, Noa Agmon, and Alfred M. Bruckstein. "A Discrete Model of Collective Marching on Rings". *Proc. of the Intl. Symposium on Autonomous Robotic Systems (DARS)*. 2021.

Michael Amir and Alfred M. Bruckstein. "Fast Uniform Dispersion of a Crash-Prone Swarm". *Robotics: Science and Systems (RSS)*. 2020.

Michael Amir and Alfred M. Bruckstein. "Minimizing Travel in the Uniform Dispersal Problem for Robotic Sensors". *Proc. of the Intl. Conf. on Distributed Autonomous Agents and MultiAgent Systems (AAMAS)*. Montreal, Canada, 2019.

Preprints

Eleanor Clifford*, **Michael Amir***, Arduin Findeis, Yiren Zhao, and Robert Mullins. *Open Problems in Preference Reconstruction*. Under review; preprint available upon request.

Academic Activities

Invited Presentations (Selected)

March 2025: Alan Turing Institute: "Learning Dynamic Constraints for Multi-Agent Controllers"

September 2023: MIT Media Lab: "Natural Algorithms"

Peer Review / Program Committees

IEEE T-RO · JAIR · ICML · ICLR · JAAMAS · ECAI · CoRL · MRS · IROS · SAC · DARS · RSS